

APHASIA COMPANION APP CONCEPT

VISION

Create a conversational companion that helps people with aphasia express themselves confidently du

CORE PRINCIPLES

1. ****User-first communication**** â Reduce frustration by prioritizing simplicity and speed over feature count.
2. ****Adaptive assistance**** â Personalize vocabulary, prompts, and interface layouts based on each personâ s speech patterns and goals.
3. ****Inclusive design**** â Provide a calm visual style, large tap targets, optional contrast modes, and cross-device sync.
4. ****Privacy-by-default**** â Keep user utterances encrypted, with clear consent before any cloud processing.

KEY USER SCENARIOS

- â€¢ ****Spontaneous chat****: The user taps a phrase starter, speaks, or types, and the app suggests likely follow-up words or sentences.
- â€¢ ****Errand support****: Before a store visit, the companion builds a quick â scriptâ with key vocabulary, reference images, and barcodes.
- â€¢ ****Care partner collaboration****: Caregivers contribute custom phrases or review progress through shared notes.
- â€¢ ****Therapy bridge****: Speech-language pathologists assign tailored practice sets and review analytics between sessions.

FEATURE OUTLINE

1. CONVERSATIONAL CANVAS

- â€¢ Live speech capture with on-device recognition tuned for aphasic speech patterns.
- â€¢ Word and phrase recommendations ranked by context, frequency, and user feedback.
- â€¢ Quick visual cards (emoji, photos, icons) to complement verbal output.

2. SMART VOCABULARY COACH

- â€¢ Progressive word banks grouped by theme and difficulty.
- â€¢ Spaced-repetition prompts blending speaking, reading, and choosing tasks.
- â€¢ Feedback loop that celebrates attempts rather than only correct answers.

3. CARE CIRCLE PORTAL

- â€¢ Shared accounts with adjustable permissions for family, friends, and clinicians.
- â€¢ Messaging layer for gentle reminders or encouragement.
- â€¢ Secure export of progress summaries for therapy documentation.

4. INTEGRATIONS

â€¢ **AI assistance**: Fine-tuned language model that understands the user's unique phrasing, suggesting clarifications or follow-up sentences.

â€¢ **Voice recognition**: Hybrid pipeline combining on-device models for privacy and optional cloud enhancement for noisy environments.

â€¢ **Wearables**: Quick prompts via smartwatch complications and haptic cues for pacing.

DATA & AI APPROACH

â€¢ Collect opt-in audio samples to adapt pronunciation models while giving users full control to delete data at any time.

â€¢ Train a small personalization layer (e.g., LoRA adapters) on-device or in a private workspace to keep the base model respectful and responsive.

â€¢ Incorporate explainabilityâ€”show why the app recommended a phrase and allow one-tap feedback to improve future suggestions.

ROADMAP SKETCH

1. **Discovery**: Interviews with people with aphasia, caregivers, and speech-language pathologists; create empathy maps.
2. **Prototype**: Figma flows and a React Native proof-of-concept focusing on the conversational canvas.
3. **Pilot**: Limited release with remote logging and weekly check-ins to observe real-world usage.
4. **Scale**: Harden infrastructure, pursue clinical validations, and add multilingual support.

SUCCESS METRICS

â€¢ Reduced average time-to-utterance during conversations.

â€¢ Increased self-reported confidence scores post-interaction.

â€¢ Daily active usage across key features (canvas, coach, care portal).

â€¢ Positive clinician feedback on therapy alignment.

ETHICAL & ACCESSIBILITY GUARDRAILS

â€¢ Transparent user agreements with plain-language summaries.

â€¢ Compatibility with screen readers and alternative input methods.

â€¢ Emergency assist mode that can call preset contacts or display medical info instantly.

NEXT STEPS

â€¢ Draft research recruitment materials with aphasia advocacy groups.

â€¢ Assemble advisory board of speech-language pathologists and lived-experience experts.

â€¢ Develop non-technical prototypes (card sorting, script rehearsal) to validate core flows before coding.